Put whole rods end to end. Leave no gaps or overlaps. You may use a length more than once. Changing the order of the same rods does not make a new way.

$3 + 4 = 7$

Problem 1: Which lengths can you build with 3-rods and 4-rods?

1

2

5

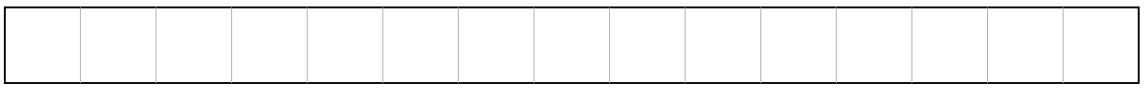
6

8

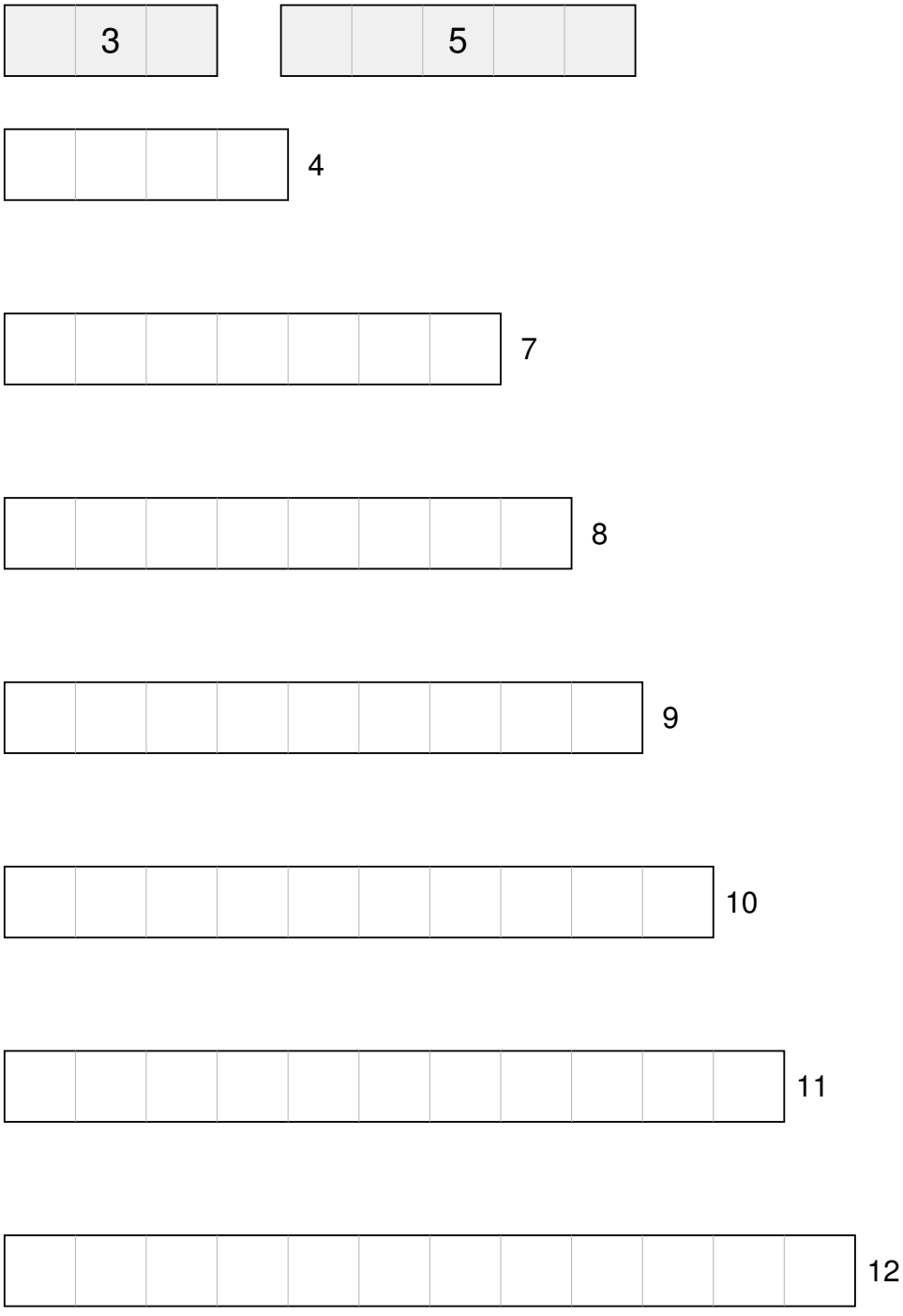
9

10

Problem 2: Find every way to make 12 and 15 with 3-rods and 4-rods.

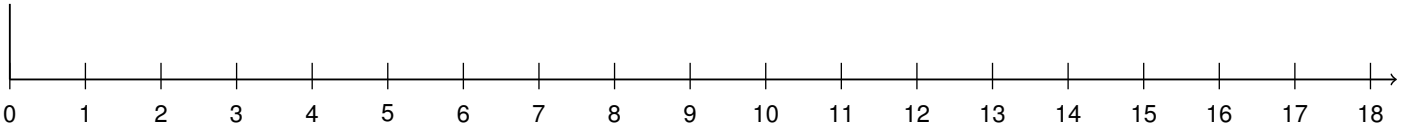


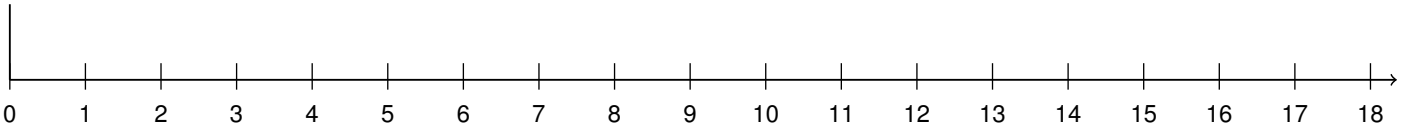
Problem 3: Use 3-rods and 5-rods. Which lengths can you build?



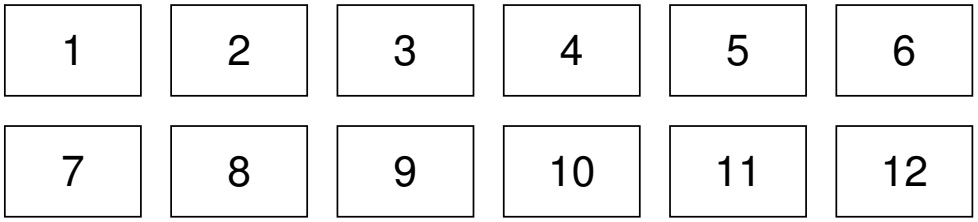
Problem 4: Use 3-rods and 4-rods to build every length from 6 through 18. Could you keep going forever?

6	7	8	9	10	11	12
13	14	15	16	17	18	

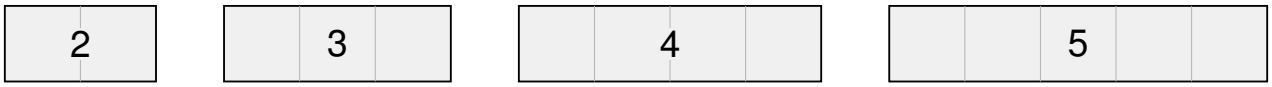




Problem 5: Use 2-rods and 4-rods to build lengths from 1 through 12. Show why some lengths cannot be built.



Problem 6: Choose two rod lengths from 2, 3, 4 and 5. Which pairs can build every target from 6 through 12?



	and		6	7	8	9	10	11	12
	and		6	7	8	9	10	11	12
	and		6	7	8	9	10	11	12
	and		6	7	8	9	10	11	12
	and		6	7	8	9	10	11	12
	and		6	7	8	9	10	11	12